Review from previously:

1. Symmetric matrices: $A^T = A$.
2. Fundamental fact about transpose: $u \cdot Av = (A^T u) \cdot v$.
3. If A is symmetric:
   (a) Two eigenvectors with distinct eigenvalues are orthogonal.
   (b) All eigenvalues are real.
4. Definition: An $n \times n$ matrix A is said to be orthogonally diagonalizable if there is a diagonal matrix D and orthogonal matrix P with $P^T = P^{-1}$ such that $A = PDP^{-1}$.
5. Theorem: A matrix is orthogonally diagonalizable if and only if it is symmetric.
   (a) Thus, A has n real eigenvalues counting multiplicity.
   (b) The eigenvectors form a complete, orthogonal basis.
6. Spectral theorem: Let A be a symmetric matrix. Fix an orthonormal basis $\{v_k\}$ of eigenvectors. Then
$$A = \lambda_1 v_1 v_1^T + \cdots + \lambda_n v_n v_n^T.$$

Definition: A **quadratic form on** $\mathbb{R}^n$ is a function (denoted by Q) whose value at x can be computed using the rule $Q(x) = x^T Ax = x \cdot Ax$, where A is a symmetric matrix.

1. Which of the following functions are quadratic forms (and which aren't)? For those that are, write the corresponding symmetric matrix.

(a) $Q(x) = x \cdot x$

(b) $Q(x) = -x_1^2 - x_2^2 + x_3^2 + x_4^2$

(c) $Q(x) = x_1x_2 + x_3x_4$

(d) $Q(x) = x_1 - x_2x_3$

(e) $Q(x) = \cos(x_1^2 - x_2^2)$

2. Consider the quadratic form $Q(x) = x_1^2 - x_2^2$. Compute the quadratic form $y \mapsto Q(Py)$ where P is the matrix below:

(a) $P = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$

(b) $P = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

3. Diagonalize the following quadratic forms (find the diagonal matrix D and the orthogonal change of variable matrix P):

(a) $Q(x) = x_1^2 - 2x_1x_2 + x_2^2$

(b) $Q(x) = 2x_1^2 - 6x_1x_2 + 2x_2^2$

(c) $Q(x) = x_1x_2 + x_3x_4$

4. For each of the quadratic forms below, what are the minimum and maximum values on the set where $\|x\| = 1$? And at what points are they achieved? And what are the saddle points and the corresponding value for Q?

(a) $Q(x) = x_1^2 - 4x_2^2$

(b) $Q(x) = 7x_1^2 + 4x_2^2 + 3x_3^2$

(c) $Q(x) = -7x_1^2 + 4x_2^2 + 3x_3^2$

(d) $Q(x) = -7x_1^2 - 4x_2^2 + 3x_3^2$

(e) $Q(x) = -7x_1^2 - 4x_2^2 - 3x_3^2$

5. Find the minimum and maximum values of the following quadratic forms on the set where $\|x\| = 1$; and give their coordinates. Also, find all saddle points with their coordinates.

(a) $Q(x) = 2x_1x_2 + 2x_2x_3 + 2x_3x_1$

(b) $Q(x) = 2x_1x_2 - 2x_2x_3$

(c) $Q(x) = 4x_1x_2 - x_1^2 - x_2^2$